

Kali Purple

Group 7

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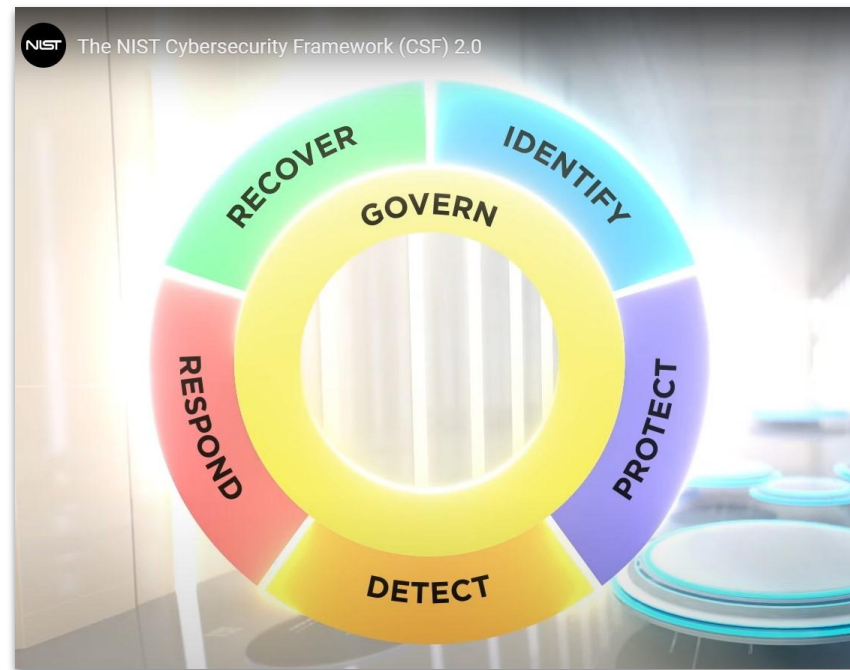
What is Kali Purple?

- A distribution of Kali Linux for defensive security
- Suite of 100+ defensive tools
- **Think: Red + Blue = Purple!**
- Platform aligned to the 5 NIST CSF Domains



NIST CSF and Kali Purple Tools

1. **Identify** - Greenbone Vulnerability Management
2. **Protect**: pfSense, CyberChef
3. **Detect**: Snort or Wireshark
4. **Respond**: TheHive
5. **Recover**: BackBox





Why Kali Purple?

- Allows for Red + Blue Team collaboration
- Open source
- Learning & Practicing Tool
- SOC-In-A-Box

SOC-In-A-Box

- Can be a complete SOC - includes offensive and defensive tools
- Efficient and Simple
- Easy incident response





Tools

Kali Autopilot

Features

- Pre-installed on Kali Purple
- Features a GUI
- Python script

Uses

- Red Team – attack simulation
- Purple Team – attack scenario training
- Penetration Testing – automate repetitive tasks



Kali Autopilot - Automated Attack Generator

Attack Scripts
juice-shop
dvwa
soc-200
soc-201

Variables for: soc-201

	Name	Value
1	subnet	192.168.253.0/24
2	js_ip	192.168.253.2
3	js_port	3000
4	purple_ip	192.168.253.3
5	appsrv	192.168.253.4
6	web01	192.168.253.6

Clear Insert Remove

Settings
Delay: 5 -- 20
Interface: eth0
API Port: 443
☒ Always insert delay
☐ Debug
☐ Set IP Addresses
☒ Loop

Scripted Attack Sequence

	Action	Refer	Prompt	Command
1	STAGE	1		Scanning
2	OS			nmap -s {subnet}
3	OS			nmap -PS -sV {js_ip}
4	STAGE	2		Fingerprinting web server
5	OS			nmap -sV {js_ip} -p {js_port}
6	OS			curl -i http://{js_ip}:{js_port}
7	OS			wget -O /dev/null http://{js_ip}:{js_port}/robots.txt
8	OS			wget -O /dev/null http://{js_ip}:{js_port}/well-known/security.txt
9	STAGE	3		Web content scanning
10	OSB			dirb http://{js_ip}:{js_port}/ /usr/share/wordlists/dirb/common.txt
11	STAGE	4		Bruteforce jmckenzie password
12	OS			wfuzz -c -w /usr/share/wordlists/dirb/others/best1050.txt -d \"en
13	STAGE	5		ssh login as jmckenzie
14	SSH	1		{appsrv},22,'jmckenzie','paramac'
15	COMMAND	1		worklog -a 5 P681-2
16	COMMAND	1		exit

Clear Insert Remove Generate

- Create or upload attack scripts
- Modify variables (define hostnames and associated IP addresses, subnets, ports, etc.)
- Modify delays between scripts, loop through scripts, etc.
- Set the sequence of attacks - action-based scripting

Greenbone Vulnerability Manager

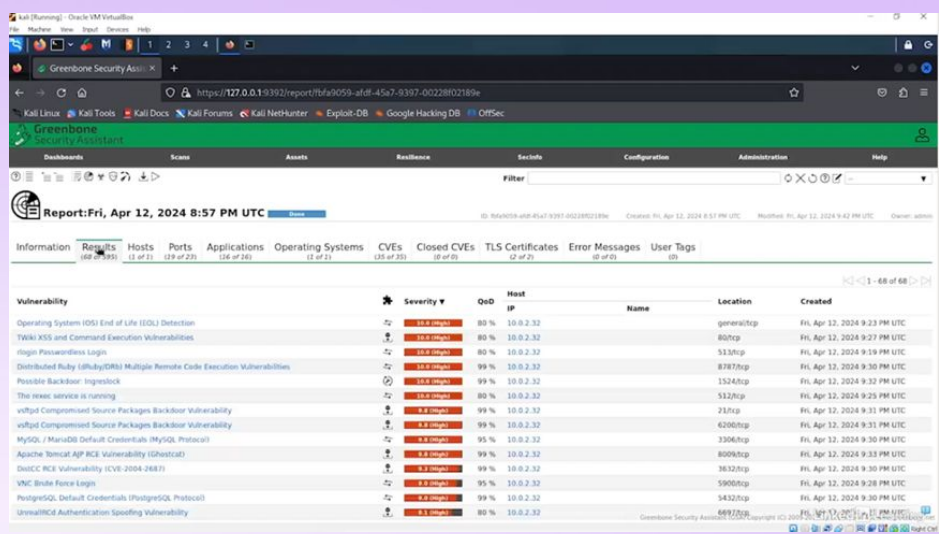
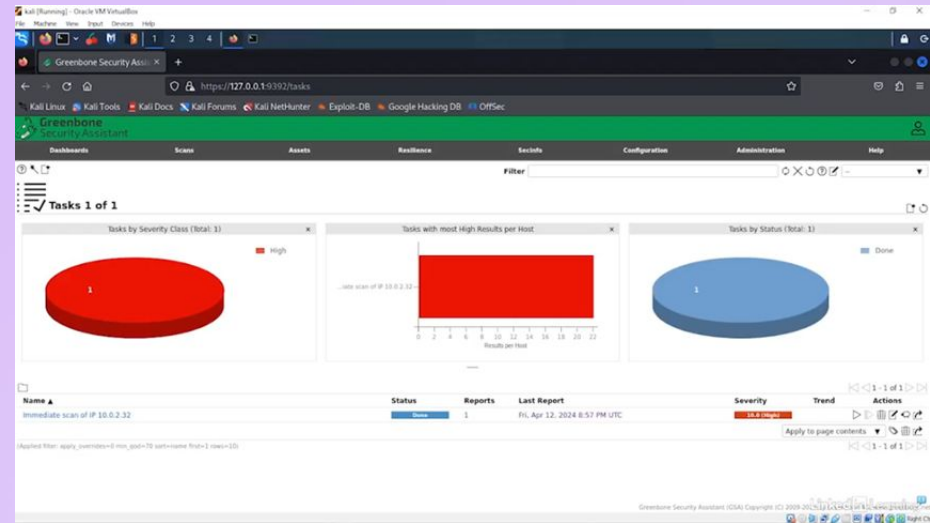
Brief History

- 2005 – Nessus discontinued open-source
- 2006 – Open Vulnerability Assessment System (OpenVAS) created
- 2008 – Greenbone AG founded in Germany to make OpenVAS better

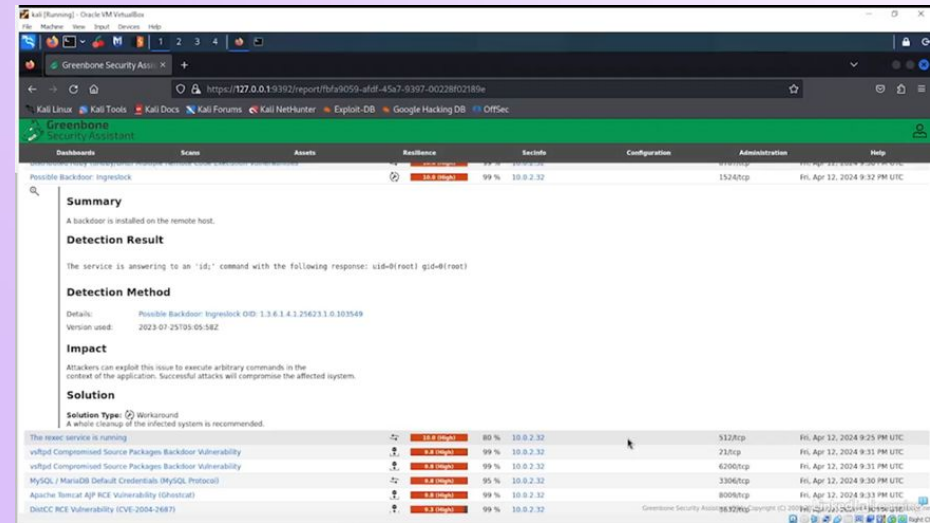
Features:

- GUI
- Over 180,000 automated vulnerability tests
- Based on nmap scanner

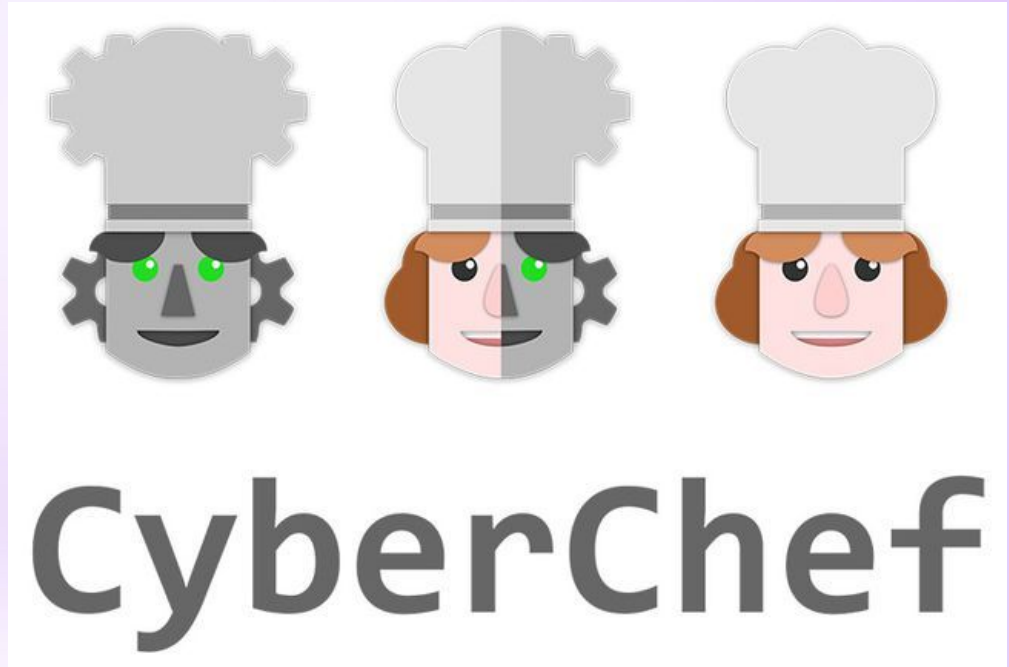




- Wizard tool
- Runs scans
- Creates reports
 - Summary
 - Solutions



- Web app
- Designed to be used without being too complex
- Serverless - compatible with many preferred browsers
- “Recipe” - drag and drop operations and define arguments and options
- Specify operations, input, output, etc.



What else does it do?

- Data encoding/decoding
- Data compression/decompression
- Cryptography
- Data analysis (parsing logs, extracting data)
- Networking (parsing IP addresses, subnet calculations)
- File operations (reading headers/footers, converting timestamps, extracting embedded files)
- “Magic” operation (automatically detects what operations can be applied to input data)
- Flexible/extendable
- Client-side processing
- Etc.



Operations

Search...

Favourites

Data format

Encryption / Encoding

Public Key

Arithmetic / Logic

Networking

Language

Operations

extra

Extract IP addresses

Extract MAC addresses

Extract email addresses

Extract ID3

Extract LSB

Extract EXIF

Extract RGBA

Extract URLs

Extract dates

Extract files

Extract domains

Extract file paths

Recipe

Extract email addresses

☒ Display total

☐ Sort

☒ Unique

Delivered-To: aromarosereji@gmail.com
Received: by 2002:a92:de4f:0:b0:335:1dba:9cc3 with SMTP id e15csp0716941lr;
Fri, 26 May 2023 01:49:09 -0700 (PDT)
X-Google-Smtp-Source: ACH4ZSKG98u3Kuglw9V7fcoEkolaw+icvjzfcXbvZH6HYSKH1pGRT19dT8zhfCEfGCI68RQVYL
X-Received: by 2002:a05:6a21:594:b0:10c:7a20:6dd6 with SMTP id lxx20-20020a056a21059400b0010c7a206dd6m1228198pzb.42.1685090948889;
Fri, 26 May 2023 01:49:08 -0700 (PDT)

Output
total found: 4
aromarosereji@gmail.com
header.i@gmail2india.org
header.i@hdfcbank.net
p-233504346-aromarosereji@gmail.com@track.hdfcbank.net

Analyze email headers

Operations

extr

Extract RGBA

Extract ID3

Extract LSB

Extract EXIF

Extract URLs

Extract dates

Extract files

Extract domains

Extract file paths

Extract IP addresses

Recipe

Fork

Split delimiter
\\n

Merge delimiter
\\n\\n

☐ Ignore errors

DNS over HTTPS

Resolver
https://cloudflare-dns.co...

Request Type
A

☐ Answer Data Only

☐ Disable DNSSEC validation

google.com
thesecmaster.com

Output
"Answer": [
 {
 "name": "google.com",
 "type": 1,
 "ttl": 130,
 "data": "142.250.205.238"
 }
]

Resolve IP addresses from domain

Date / Time

Extractors

Compression

Hashing

Code tidy

Forensics

Multimedia

Other

Flow control

Recipe

Analyse hash

19dbed996b1a814658bef433bad62b03e5c59c2bf2351b793d1a5d4a5216d27e

Output
Hash length: 64
Byte length: 32
Bit length: 256

Based on the length, this hash could have been generated by one of the following hashing functions:
SHA-256
SHA3-256
BLAKE-256
ECOH-256
FSB-256
GOST
Grøstl-256
HAVAL-256
PANAMA
RIPEMD-256
Snefru

Hash analysis



Load installer components from installation media

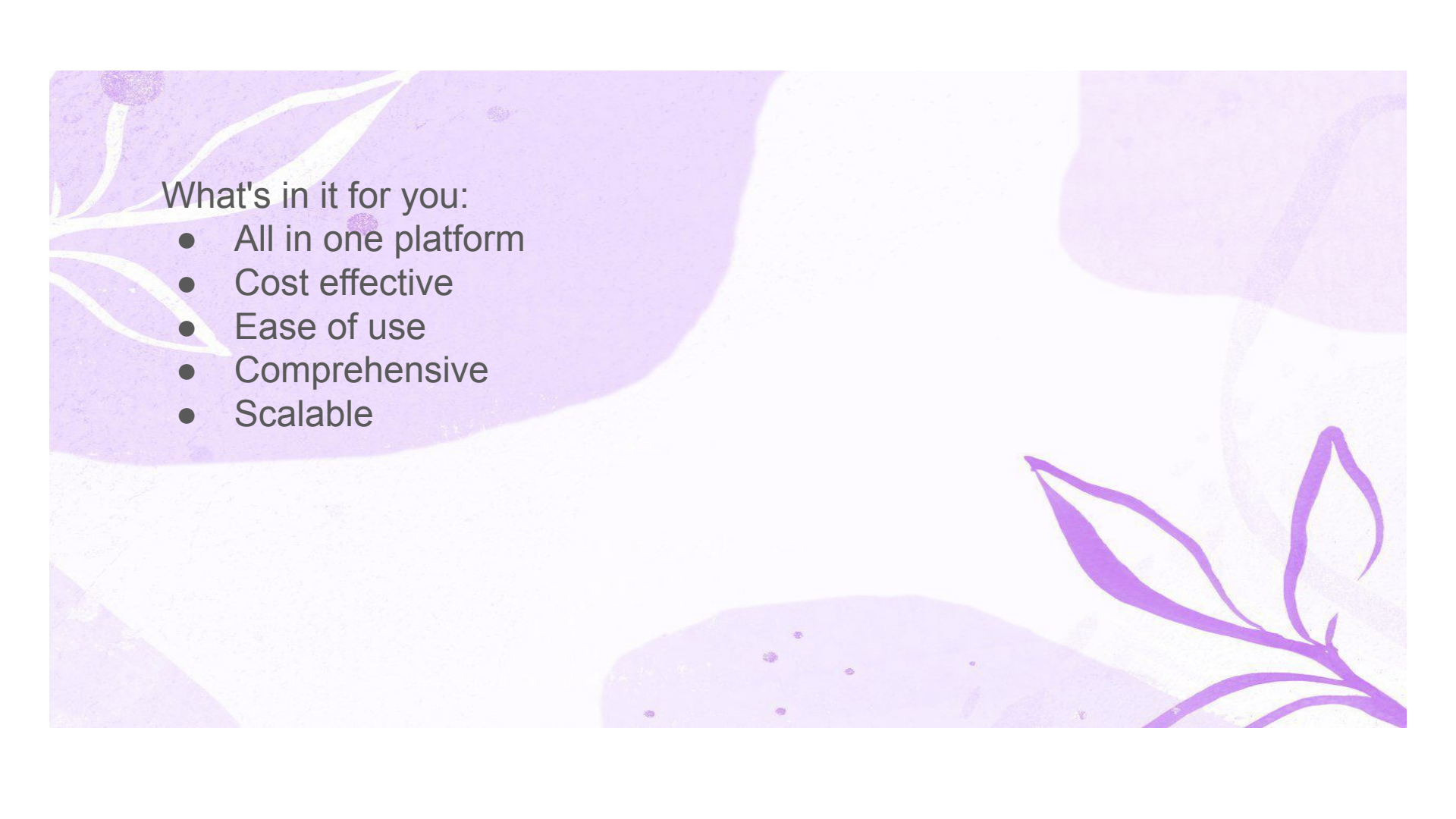
Loading additional components

Configuring simple-cdd-profiles





Say yes to the dress!!!



What's in it for you:

- All in one platform
- Cost effective
- Ease of use
- Comprehensive
- Scalable

Resources

<https://www.stationx.net/kali-purple-guide/>

<https://www.cbtnuggets.com/blog/technology/security/why-kali-linuxs-purple-distro-is-an-ethical-hackers-dream>

<https://kalilinuxtutorials.com/cyberchef-encryption-encoding-compression-data-analysis/>

<https://www.npmjs.com/package/cyberchef>

<https://theseckmaster.com/blog/exploring-cyberchef-the-cyber-swiss-army-knife-of-security-analysts-soc-analyst>

<https://blog.desdelinux.net/en/kali-linux-2023-3-arrives-with-kali-autopilot-new-tools-and-more/>

<https://www.linkedin.com/learning/kali-purple-essential-training/using-kali-autopilot-to-generate-attack-scripts?resume=false>

<https://www.greenbone.net/en/>

<https://greenbone.github.io/docs/latest/background.html>

https://wiki.gentoo.org/wiki/Greenbone_Vulnerability_Management

<https://www.linkedin.com/learning/penetration-testing-essential-training-24352676/scan-targets-with-gvm>

<https://www.nist.gov/cyberframework>